Layered System Architecture

EdrawMax-style layered view for the Finance Risk & Investment Intelligence System (inspired by AI-Based System Architecture Diagram Template). Top-to-bottom: Users/UI ? Application ? AI Agents ? Data ? Infrastructure, with Safety and Observability as cross-cutting bands.

Pause | Prev | Next | Normal | Step 1 of 7

Step 1 of 7: Users enter via the Interface layer
Analysts, risk managers, PMs, and compliance ask natural-language questions through the frontend / API client.

1 USER / INTERFACE LAYER

- **Investment Analyst** — Policy & portfolio questions
- **Risk Manager** — Thresholds & breaches
- **Portfolio Manager** — Allocation & decisions
- **Compliance** — Audit & escalation
- **Frontend / API Client** — Sprint 5+ UI + JWT

? natural-language query + auth token

2 APPLICATION LAYER (FASTAPI)

- **POST /api/v1/query** — Request / response envelope
- **Auth + RBAC** — role ? allowed_portfolio_ids
- **Admin / Health** — /health ? /metrics ? /admin
- **Pydantic Schemas** — citations ? risk ? escalation

? classified path into LangGraph

3 AI / AGENT LAYER (LANGGRAPH + LLM)

- **Router Agent** — C1?C5 ? entities ? risk level
- **Retrieval Agent** — Hybrid RAG over policies
- **Portfolio Eval** — Holdings ? concentration
- **Market Data** — Indicators ? trends
- **Risk Agent** — Scorer ? breach detect
- **Synthesis** — Tool-less, evidence only
- **Escalation Agent** — Human handoff envelope

? evidence from stores (read-only)

4 DATA / KNOWLEDGE LAYER

Policy Corpus
7 PDFs • page/section meta

pgvector
policy_chunks embeddings

finance_risk_db
5 tables • SELECT-only

SQL Templates
18 named parameterized

Session Memory
last N turns • entities

finance_auth_db
Users / JWT (separate)

? runtime platform

5 INFRASTRUCTURE LAYER

PostgreSQL 16
+ pgvector extension

OpenAI-compatible LLM
router / agent / judge tiers

Docker Compose
local • CI • prod stubs

Eval Harness
50 golden + attack suite

X1 SAFETY & GOVERNANCE (CROSS-CUTTING)

Input Guardrail
override / bypass / injection

E1•E8 Engine
deterministic hard gates

Output Validator
citations • no advice

RBAC Filter
0 unauthorized access

X2 OBSERVABILITY & COST (CROSS-CUTTING)

structlog JSON
per-request audit

OpenTelemetry
span per agent node

Prometheus
/metrics

Token / Cost Meter
per category spend

Colour bands: blue = UI • teal = Application • purple = AI agents • amber = Data • slate = Infra • coral = Safety • sky = Observability

1 **Users enter via the Interface layer**
Analysts, risk managers, PMs, and compliance ask natural-language questions through the frontend / API client.

2

Application layer authenticates the request
FastAPI accepts POST /api/v1/query, resolves JWT ? role + allowed_portfolio_ids, and stamps trace_id.

3

Safety runs before any expensive AI work
Input guardrail blocks override / bypass / injection. Escalation rules E1–E8 are deterministic hard gates.

4

AI layer routes and fans out agents
Router classifies C1–C5. Retrieval, Portfolio Eval, Market Data, and Risk collaborate through GraphState — not free chat.

5

Data layer supplies grounded evidence
pgvector holds 7 policy PDFs; finance_risk_db answers structured lookups via named SQL templates only.

6

Synthesis + validation produce the answer
Tool-less Synthesis may only use the evidence bundle. Output validator enforces citations, no advice, disclaimer.

7

Escalation or observability close the loop
High-risk paths hand off to a human with full context. Every node emits logs, traces, metrics, and cost.

Source: planning/03_Initial_System_Blueprint.md — planning/06_High_Level_Architecture.md — style inspired by EdrawMax AI system architecture template